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PACL-CDR: Popularity-Adaptive Contrastive Learning for Cross-Domain Recommendation

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***Abstract*—Cross-domain recommendation (CDR) has garnered significant attention for its potential to mitigate data sparsity and cold-start issues by facilitating knowledge transfer between source and target domains. While recent contrastive CDR methods have attempted to address information heterogeneity and behavioral discrepancies via positive and negative pair construction, two persistent challenges remain. (1) Popularity bias, where unpopular items are frequently misclassified as negative samples, resulting in an overemphasis on popular items and the neglect of unpopular items. (2) Over-reliance on overlapping users, which restricts the construction of positive and negative pairs, hinders the modeling of non-overlapping user preferences, and results in incomplete or biased knowledge transfer.**

To address these challenges, we propose a novel cross-domain recommendation framework, PACL-CDR, which incorporates a Debiasing Popularity Module (DPM) and an Intent Enhancement Module (IEM). (1) DPM leverages alignment and contrastive learning strategies to balance the feature representations of head (popular) and tail (unpopular) items, and further enhances semantic consistency between them by constructing cross-domain item tags. (2) IEM transfers cross-domain knowledge through latent user intents, thereby reducing its reliance on limited overlapping-user correspondences. Extensive experiments on four benchmark cross-domain recommendation tasks demonstrate the competitive overall performance of PACL-CDR, which achieves the strongest mean results in most evaluated settings. The source code is publicly available at https://github.com/Alpacanm/PACL-CDR.

***Index Terms*—Cross-domain Recommendation, Popularity Bias, Intent Modeling, Contrastive Learning**

I. Introduction

Cross-Domain Recommendation (CDR) leverages source-domain knowledge to alleviate data sparsity [1], [2] and cold-start issues [3]–[5], which hinder accurate preference modeling in the target domain. Traditional CDR methods, including matrix factorization (MF) [6], [7] and multilayer perceptron (MLP) models [8], exploit cross-domain relatedness through shared representations or joint learning. For instance, CoNet [9] constructs unified cross-domain feature representations through bidirectional mappings implemented by cross-connection units between the source and target domain networks. Fundamentally, these traditional models operate on the premise that users exhibit consistent behavioral patterns or share common features across domains. However, this assumption is frequently undermined by information heterogeneity and the prevalence of domain-specific user behaviors.

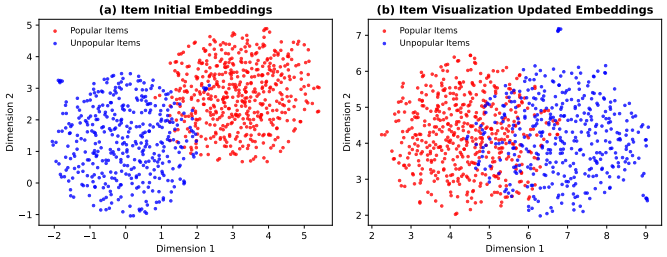


Fig. 1: Items in the sport&phone cross-domain scenario are categorized as popular or unpopular based on predefined interaction thresholds. In (a), popular and unpopular items show a clear separation in the vector space. In (b), the alignment and contrastive strategies in DPM reduce the observed separation and produce greater integration between the two groups.

To address these challenges, recent CDR methods have increasingly adopted contrastive learning (CL) [10], [11] to learn more transferable cross-domain representations. By constructing positive and negative samples, contrastive learning enhances the alignment of learned representations. To mitigate domain heterogeneity, CL methods aim to align shared information between the source and target domains while filtering out irrelevant, domain-specific noise. For example, DisCo [12] leverages intent-level contrastive learning to enhance user similarity within each domain. To address inconsistencies in user behavior across domains, CL methods instead focus on aligning the preferences of the same user across different domains while distinguishing them from those of other users. For example, the CTT model [13] constructs positive samples from a user's cross-domain interactions and negative samples from interactions involving other users and items selected at random. Through the contrastive strategy, the model effectively captures both shared and domain-specific behavioral patterns.

Despite the advancements achieved by Contrastive Learning-based Cross-Domain Recommendation (CL-CDR), several critical challenges remain, limiting its practical applicability. The present study identifies and analyzes two main challenges, highlighting their underlying causes and implications:

Challenge 1: Neglect of Popularity Bias. Popularity bias leads models to over-recommend popular items while learning inadequate representations for unpopular items with sparse interactions. In CL-CDR [14], [15], the process of constructing positive and negative pairs exacerbates popularity bias: unpopular items are frequently selected as negative samples due to their sparse features, while popular items are

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disproportionately chosen as positive samples. During model training, the resulting imbalance strengthens the representation of positive samples while further marginalizing negative samples, especially unpopular items. Moreover, as shown in Fig. 1(a), popular and unpopular items differ not only in their interaction features but also exhibit a significant separation in the embedding space. The representational separation weakens the alignment and consistency between the embeddings of popular and unpopular items, thereby limiting the model's ability to accurately capture user preferences.

Challenge 2: Over-reliance on Sparse Overlapping Users. In cross-domain recommendation, overlapping users are commonly used to align the source and target domains and transfer preference knowledge across them. However, the number of overlapping users is often limited, and sparsity undermines the strength of cross-domain connections. Moreover, in CL-CDR [14], [16], the behaviors of the same user across different domains are typically treated as positive samples, while those of different users are randomly selected as negative samples. Such a sampling strategy causes the model to overemphasize the preferences of overlapping users while neglecting those of non-overlapping users. As a result, the model provides high-quality recommendations for overlapping users but performs poorly for non-overlapping ones, thereby reducing the generalization capability across users.

To address the identified challenges, we propose PACL-CDR, a novel model with two primary components: a Debiasing Popularity Module (DPM) and an Intent Enhancement Module (IEM). First, to address **Challenge 1**, DPM integrates alignment and contrastive strategies to reduce cross-domain popularity bias. Specifically, the alignment strategy leverages popular items as supervisory signals to improve the feature representations of unpopular items. Simultaneously, the contrastive strategy introduces a popularity-weighted loss to reduce the spatial gap between popular and unpopular items in the embedding space, as illustrated in Fig. 1(b). Furthermore, DPM leverages tag-driven item associations to promote feature sharing and enhance semantic consistency across domains. Second, to address **Challenge 2**, IEM is designed to reduce the reliance on sparse overlapping users. Rather than depending solely on user overlap for knowledge transfer, IEM clusters users with similar intents, shifting the focus to their shared behavioral patterns. This intent-based modeling strengthens cross-domain connections and captures deeper, abstract preferences. By integrating DPM and IEM, PACL-CDR jointly performs popularity-aware item learning and intent-level cross-domain transfer. In this paper, we make the following contributions:

- *Conceptual:* We systematically analyze the amplification of item popularity bias in cross-domain recommendation and introduce a combined alignment and contrastive framework to bridge the representational gap between popular and unpopular items.
- *Methodological:* We propose **PACL-CDR**, a popularity-adaptive contrastive learning framework in which DPM explicitly redistributes learning signals according to item popularity and IEM provides an intent-level transfer pathway designed to reduce sensitivity to scarce overlapping-user correspondences.
- *Experimental:* Experiments on four real-world datasets show that **PACL-CDR** achieves the strongest mean ranking performance in most evaluated settings. Targeted tail-exposure and cross-domain link-severing analyses provide additional evidence for the intended roles of DPM and IEM within the evaluated protocols.

II. METHODOLOGY

As shown in Fig. 2, the overall framework of our proposed PACL-CDR: (1) **Debiasing Popularity Module (DPM)** first refines cross-domain item embeddings via alignment and contrastive strategies, using popular items to enrich and stabilize the representations of unpopular items. Building on these unified embeddings, DPM then constructs item tags that encode popularity-related semantics, so that unpopular items can borrow semantic and popularity cues from popular ones, thereby alleviating the bias toward frequently interacted items. (2) **Intent Enhancement Module (IEM)** focuses on modeling user preferences at a deeper level by emphasizing shared user intents. Some procedures are consistent across the source and target domains, and we take the **source domain as an example** for illustration. At a high level, DPM regulates item-level learning signals according to popularity, while IEM transfers cross-domain information through latent intents rather than relying solely on sparse overlapping-user correspondences.

A. *GNN Encoder*

We propose a GNN-based encoder to learn robust representations by capturing high-order collaborative signals. To mitigate data sparsity and noise, the module integrates multi-layer message passing with residual connections to prevent oversmoothing, alongside a self-supervised denoising mechanism that dynamically filters spurious interactions via a structure-aware contrastive objective.

*1) **Enriching Embeddings via Graph Propagation**:* Prior work [17] highlights that sparse interaction data limits the discovery of latent relationships, particularly for cold-start users. To address this, we employ GNNs to propagate information over the user–item graph, capturing multi-hop collaborative signals. We model user–item relations as a bipartite graph with interaction matrix $A_{uv} \in \mathbb{R}^{U \times V}$. At each layer l, embeddings are updated as:

$$E_u^l = \sigma\big(A_{uv}\, E_v^{l-1} W_u^l\big), \tag{1}$$

$$E_v^l = \sigma\big(A_{uv}^T\, E_u^{l-1} W_v^l\big), \tag{2}$$

where W_u^l, W_v^l are learnable matrices and $\sigma(\cdot)$ is a non-linear activation.

Deep propagation risks oversmoothing, causing embeddings to lose node-specific characteristics. We therefore introduce a residual connection from the initial embeddings to preserve original features:

$$E_u = E_u^0 + \mathrm{Mean}\left(\{E_u^0, \ldots, E_u^L\}\right), \tag{3}$$

$$E_v = E_v^0 + \mathrm{Mean}\left(\{E_v^0, \ldots, E_v^L\}\right). \tag{4}$$

The resulting embeddings E_u and E_v are used for all downstream tasks.

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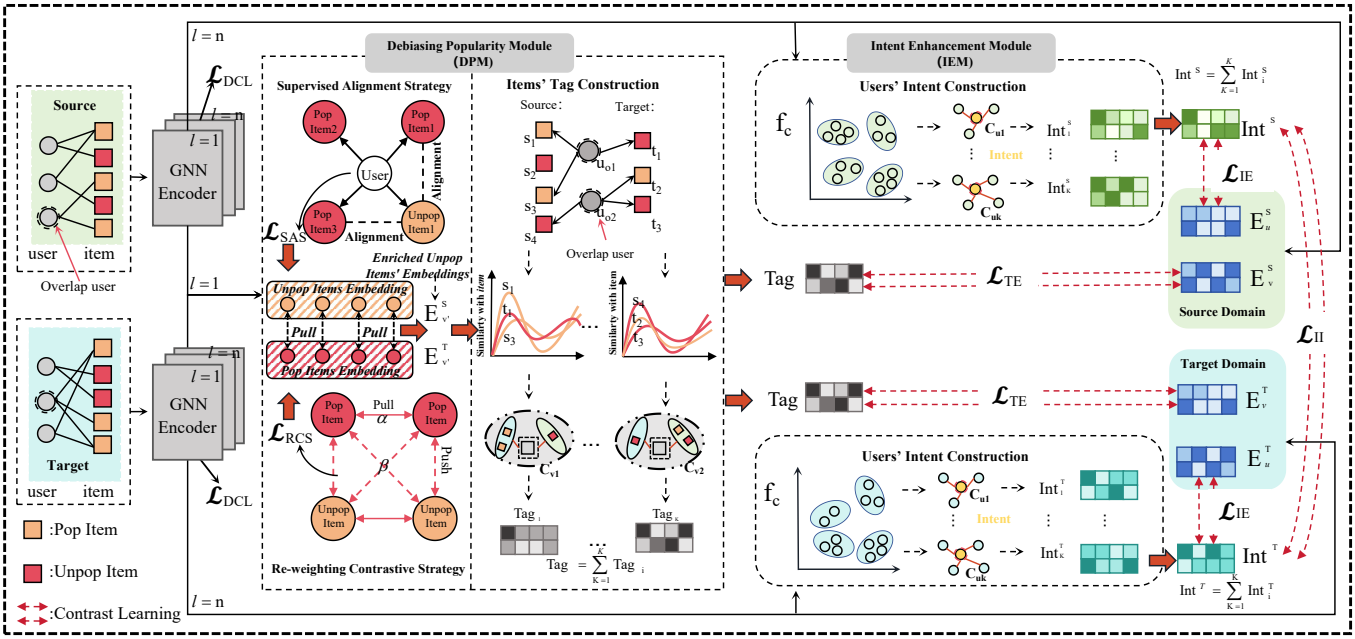


Fig. 2: The overall framework of our proposed PACL-CDR.

*2) **Denoising Contrastive Loss** L_{DCL}:* Real-world interaction graphs inevitably contain noisy edges (e.g., accidental clicks), which propagate misleading signals and degrade feature quality. To enhance robustness, we introduce an auxiliary denoising contrastive objective. The core idea is to generate a denoised view of the adjacency matrix and encourage it to be structurally consistent with the original graph, so that the encoder learns to suppress noise while preserving the main interaction patterns.

Following [18], we first estimate the importance $\pi_{i,j}$ of each edge based on the embeddings of the connected nodes:

$$\pi_{i,j} = \sigma\big(W_p^\top [E_{u_i}^{l-1}, E_{v_j}^{l-1}]\big), \tag{5}$$

where W_p is a learnable weight vector and $[\cdot,\cdot]$ denotes concatenation. To filter noise while maintaining differentiability, we generate a soft mask M using the Gumbel–Softmax trick:

$$M = \sigma\left(\frac{\log \pi_{i,j} + g}{\tau}\right), \tag{6}$$

where $g \sim \text{Gumbel}(0,1)$ and τ is the temperature for mask generation. The denoised adjacency matrix is then defined as:

$$A'_{uv} = A_{uv} \odot M. \tag{7}$$

To explicitly regularize the denoising process, we treat the original adjacency matrix A_{uv} and its denoised counterpart A'_{uv} as a pair of positive samples in a contrastive framework. Following the same InfoNCE form as in the embedding-level contrastive loss, we define the denoising contrastive loss as:

$$L_{\text{DCL}} = -\frac{1}{U}\sum_{i=1}^{U} \log \frac{\exp\big(s(A_{u_i v}, A'_{u_i v})/\tau_c\big)}{\sum_{j=1}^{U} \exp\big(s(A_{u_i v}, A'_{u_j v})/\tau_c\big)}, \tag{8}$$

where $s(\cdot,\cdot)$ denotes the cosine similarity. The L_{DCL} encourages each original adjacency $A_{u_i v}$ to be most similar to its own denoised version $A'_{u_i v}$, while being dissimilar to the denoised structures $\{A'_{u_j v}\}_{j\neq i}$, thereby guiding the denoising module to preserve instance-specific interaction patterns and suppress noisy edges.

B. *Debiasing Popularity Module (DPM)*

To address popularity-induced representation imbalance in recommender systems [19]–[21], DPM improves tail-item representations through a **Supervised Alignment Strategy**, a **Re-weighting Contrastive Strategy**, and **Item Tag Construction**. By aligning tail items with head items and restructuring the embedding space, DPM regularizes the distribution used to generate semantic tags; this design does not guarantee unbiased representations. To operationalize this module, we first organize items based on their alignment with each user's interests. For any user u, we refer to the set of items they interact with as V_u:

$$V_u = \{v \mid v \in V \text{ and } A_{uv} = 1\}. \tag{9}$$

To quantify the popularity of each item i, we follow prior studies [22], [23] and compute $p(i)$, the proportion of item i appearing in the training dataset. Specifically, $p(i)$ represents the proportion of users who have interacted with item i among the total number of users, capturing how often the item appears in user interactions. Based on $p(i)$, we divide V_u into two groups: the popular item group V_u^{pop} and the unpopular item group V_u^{unpop}:

$$V_u = V_u^{pop} \cup V_u^{unpop}, \tag{10}$$

$$\forall i \in V_u^{pop} \text{ and } i' \in V_u^{unpop},\ p(i) > p(i'), \tag{11}$$

where V_u^{pop} and V_u^{unpop} are disjoint sets, and every item in the popular group has higher popularity than any item in the unpopular group.

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*1) **Supervised Alignment Strategy**:* Items interacted with by the same user often reflect related preferences. Based on this observation, we construct cross-popularity item pairs within each user's interaction history and define the following supervised alignment loss:

$$L_{\text{SAS}} = \sum_{u \in U} \frac{1}{|V_u|} \sum_{\substack{i \in V_u^{\text{pop}} \\ j \in V_u^{\text{unpop}}}} \|E_{v_i} - E_{v_j}\|^2, \tag{12}$$

where V_u^{pop} and V_u^{unpop} denote the sets of popular and unpopular items interacted with by user u in domain S, respectively, and E_{v_i} denotes the representation of item i.

By minimizing L_{SAS}, popular and unpopular items associated with the same user are encouraged to remain close in the embedding space. This user-conditioned constraint provides additional relational supervision for sparsely interacted items and reduces the representation separation between different popularity groups.

*2) **Re-weighting Contrastive Strategy**:* To account for popularity imbalance in item-side contrastive learning, we construct two augmented views of each item embedding as $h_i' = E_{v_i} + \Delta_i'$ and $h_i'' = E_{v_i} + \Delta_i''$, where Δ_i' and Δ_i'' are normalized random noise vectors. The items in the current batch $\mathcal{B}$ are then divided into popular and unpopular sets, denoted by V^{pop} and V^{unpop}, respectively. Based on this partition, we define separate contrastive losses for the two popularity groups and re-weight their cross-popularity negatives.

For popular items, the contrastive loss incorporates a hyperparameter β to control the penalty strength of unpopular items serving as negative samples:

$$L_{\text{pop}} = -\sum_{i \in V_u^{\text{pop}}} \log \frac{\exp\left(h_i' \cdot h_i''/\tau\right)}{\sum_{j \in V_u^{\text{pop}}} \exp\left(h_i' \cdot h_j''/\tau\right) + \beta \sum_{k \in V_u^{\text{unpop}}} \exp\left(h_i' \cdot h_k''/\tau\right)}. \tag{13}$$

Similarly, for unpopular items, we adjust the influence of popular items in the denominator:

$$L_{\text{unpop}} = -\sum_{i \in V_u^{\text{unpop}}} \log \frac{\exp\left(h_i' \cdot h_i''/\tau\right)}{\sum_{j \in V_u^{\text{unpop}}} \exp\left(h_i' \cdot h_j''/\tau\right) + \beta \sum_{k \in V_u^{\text{pop}}} \exp\left(h_i' \cdot h_k''/\tau\right)}. \tag{14}$$

The final item-side contrastive loss is a weighted combination controlled by α:

$$L_{\text{RCS}} = \alpha \cdot L_{\text{pop}} + (1 - \alpha) \cdot L_{\text{unpop}}, \tag{15}$$

where τ is the temperature parameter. The hyperparameter α balances the focus between popular and unpopular positive samples, while β re-weights the negative samples to prevent pushing diverse items too far apart in the embedding space. Therefore, L_{RCS} allows the two popularity groups to contribute differently to contrastive optimization, rather than assigning identical roles to all anchors and negatives. With appropriate weighting, this objective is intended to balance popularity-related learning signals and mitigate representation separation between popular and unpopular items. This provides an optimization-level rationale for alleviating popularity bias, rather than a formal guarantee of its elimination.

*3) **Item Tag Construction**:* In cross-domain recommendation, interaction data typically follows a long-tailed distribution, causing models to overfit popular items while under-representing unpopular ones. To explicitly incorporate popularity information and alleviate this bias, we propose an item tag construction mechanism based on global semantic prototypes. The core idea is to map item pairs into a shared latent space where common semantic features are highlighted based on their co-occurrence confidence.

Before constructing tags, we obtain the popularity-debiased and structure-enhanced item embeddings from the source and target domains. After applying the supervised alignment loss $\mathcal{L}_{\text{SAS}}$ and the re-weighted contrastive loss $\mathcal{L}_{\text{RCS}}$, the items are encoded as:

$$E_{v'}^S \in \mathbb{R}^{V \times d}, \quad E_{v'}^T \in \mathbb{R}^{V \times d}, \tag{16}$$

where $E_{v'}^S$ and $E_{v'}^T$ denote the optimized representations of items v_S and v_T, respectively. These embeddings encapsulate the intrinsic properties of items within their domains.

For source-target item pairs connected through overlapping users, we compute their co-occurrence similarity $\text{Sim}(v_S, v_T)$ to measure the strength of their cross-domain association:

$$\text{Sim}(v_S, v_T) = \frac{\sum_{u \in \mathcal{U}'} \phi(u, v_S) \cdot \phi(u, v_T)}{\sqrt{\sum_{u \in \mathcal{U}'} \phi(u, v_S)^2} \cdot \sqrt{\sum_{u \in \mathcal{U}'} \phi(u, v_T)^2}}, \tag{17}$$

where $\mathcal{U}'$ is the set of overlapping users. Notably, this similarity computation delineates the explicit boundary of overlapping user utilization within the DPM; their interaction records are exclusively leveraged here to bridge cross-domain item pairs, rather than being directly encoded into the prompt representations. Here, $\phi(u, v)$ is the binary interaction indicator.

Instead of treating items in isolation, we aim to extract the *common semantic features* activated by the co-occurrence of the pair (v_S, v_T). We generate a unified representation $\tilde{E}$ by fusing the paired embeddings:

$$\tilde{E} = \text{Sim}(v_S, v_T) \cdot \left(E_{v'}^S \odot E_{v'}^T\right), \tag{18}$$

where the element-wise product $\odot$ acts as a **soft feature intersection**, highlighting latent dimensions that are active in both domains. Crucially, the similarity score $\text{Sim}(v_S, v_T)$ serves as a **confidence gate**. It ensures that the fused features are emphasized only when there is strong evidence of co-consumption (high similarity), effectively suppressing noise from accidental or weak correlations.

To encode these contexts into discrete semantics, we introduce a set of global latent prototypes $\text{C}_v = \{c_{v1}, \ldots, c_{vk}\}$, parameterized by matrix $M_v = [c_{v1}, \ldots, c_{vk}] \in \mathbb{R}^{d \times K}$. These prototypes serve as canonical centers for popularity-aware semantics. We compute the association of the proxy representation with these prototypes:

$$\gamma_v = \text{softmax}\left(\tilde{E} \cdot M_v\right), \tag{19}$$

where γ_v indicates the probability that the current item pair belongs to the k-th semantic cluster.

Finally, we generate K tag representations for the item pair. The k-th tag is projected from the shared proxy embedding, guided by the specific prototype:

$$\text{Tag}_k = \sigma\left(\tilde{E} W_k + \gamma_v \cdot \mathbf{b}_k\right), \quad k \in \{1, \ldots, K\}, \tag{20}$$

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where $W_k \in \mathbb{R}^{d\times d}$ and $\mathbf{b}_k \in \mathbb{R}^d$ are learnable parameters for the k-th tag. Through this mechanism, unpopular items can inherit rich semantic tags. If an unpopular item v_T co-occurs with a popular item v_S, the fused $\tilde{E}$ will be enhanced by the strong features of v_S weighted by their similarity, effectively allowing the unpopular item to align with stable global prototypes.

C. *Intent Enhancement Module (IEM)*

Traditional recommendation methods typically represent a user with a single embedding vector E_u, which entangles multifaceted preferences (e.g., price sensitivity, functional needs, or aesthetic interests) into a unified representation. In cross-domain scenarios, such entangled representations generalize poorly because the dominance of specific intents varies significantly across domains. To address this limitation, the Intent Enhancement Module (IEM) decomposes each user embedding into K intent-specific high-dimensional representations, shifting the focus from isolated personalized signals to global behavioral trends governed by shared latent motivations. To guide this disentanglement process with semantic consistency, we introduce a set of global intent prototypes $\mathrm{C}_u = \{c_{u1}, c_{u2}, \ldots, c_{uk}\}$, where each $c_{uk} \in \mathbb{R}^d$ represents a canonical pattern of the k-th latent intent. This is analogous to clustering user behaviors in the embedding space and using the cluster centroids as global references. We parameterize these prototypes with a learnable matrix

$$M_u = \big[c_{u1}, \ldots, c_{uk}\big], \tag{21}$$

and compute the intent association scores for user u as

$$\gamma_u = \mathrm{softmax}\big(E_u \cdot M_u\big), \tag{22}$$

where the k-th element γ_u measures how strongly user u is related to the k-th intent prototype. Leveraging these global prototypes as semantic anchors, we generate K intent-specific high-dimensional embeddings for user u. The representation of the k-th intent, denoted as Int_k, is formulated as:

$$\mathrm{Int}_k = \sigma\big(E_u W_k + \gamma_u\, b_k\big), \quad k \in \{1, 2, \ldots, K\}, \tag{23}$$

where $W_k \in \mathbb{R}^{d\times d}$ and $\mathbf{b}_k \in \mathbb{R}^d$ are the learnable projection matrix and bias vector for the k-th intent, and $\sigma(\cdot)$ is the activation function. Here, W_k specifies how the generic user embedding E_u is transformed when focusing on the k-th intent, while γ_u controls the strength of the k-th intent component in Int_k. By explicitly modeling multiple intent-specific user embeddings, the IEM alleviates the entanglement of heterogeneous preferences in the single-vector representation E_u. Different behavioral signals (e.g., price-driven clicks vs. function-driven purchases) can be selectively captured by different Int_k, which allows the model to focus on the most relevant intent.

D. *Triple Contrastive Learning*

1) ***Contrastive Loss for Intent Embeddings*** L_{IE}*:* To make the global user embedding E_u consistent with the disentangled intent space, we aggregate the K intent representations $\{\mathrm{Int}_1, \ldots, \mathrm{Int}_K\}$ into a unified intent embedding:

$$\mathrm{Int} = \sum_{k=1}^{K} \mathrm{Int}_k. \tag{24}$$

We then apply a contrastive objective to pull each user's global embedding E_u close to their own intent embedding Int_u, while pushing it away from other users' intents:

$$L_{\mathrm{IE}} = -\frac{1}{U}\sum_{i=1}^{U} \log \frac{\exp\big(s(E_{u_i}, \mathrm{Int}_{u_i})/\tau_m\big)}{\sum_{j=1}^{U}\exp\big(s(E_{u_i}, \mathrm{Int}_{u_j})/\tau_m\big)}, \tag{25}$$

where $s(\cdot,\cdot)$ denotes the cosine similarity and τ_m is the temperature hyperparameter. The L_{IE} encourages E_u to encode the user's multi-intent preference in a coherent way, rather than behaving as a simple ID embedding.

2) ***Contrastive Loss for Tag Embeddings*** L_{TE}*:* For each item v, we obtain K latent tag embeddings $\{\mathrm{Tag}_1, \ldots, \mathrm{Tag}_K\}$ and aggregate them into a unified tag representation:

$$\mathrm{Tag} = \sum_{k=1}^{K} \mathrm{Tag}_k. \tag{26}$$

We then use a contrastive objective to align the original item embedding E_v with its aggregated tag embedding Tag, and to push it away from the tags of other items:

$$L_{\mathrm{TE}} = -\frac{1}{V}\sum_{i=1}^{V} \log \frac{\exp\big(s(E_{v_i}, \mathrm{Tag}_{v_i})/\tau_m\big)}{\sum_{j=1}^{V}\exp\big(s(E_{v_i}, \mathrm{Tag}_{v_j})/\tau_m\big)}, \tag{27}$$

where $s(\cdot,\cdot)$ denotes the cosine similarity, and τ_m is the temperature hyperparameter. The L_{TE} encourages E_v to be consistent with its multi-tag semantics, making item embeddings more informative and semantically structured.

3) ***Cross-Domain Intent Contrastive Loss*** L_{II}*:* The user intent representations learned from the source and target domains often exhibit significant differences due to varying behavior patterns across domains. For example, user behaviors in different domains (e.g., browsing products versus searching for movies) may lead to distinct representations, making direct alignment challenging. We adopt contrastive learning as an alignment method. By constructing positive and negative sample pairs, the method maximizes the similarity of positive pairs while minimizing the similarity of negative pairs, enabling robust representation learning:

$$L_{\mathrm{II}} = -\frac{1}{U}\sum_{i=1}^{U} \log \frac{\exp\big(s(Int^S_{u_i}, Int^T_{u_i})/\tau_m\big)}{\sum_{j=1}^{U}\exp\big(s(Int^S_{u_i}, Int^T_{u_j})/\tau_m\big)}, \tag{28}$$

where $Int^S_{u_i}$ and $Int^T_{u_i}$ represent the intent representations of overlapping user u_i in the source domain and the target domain, respectively. The similarity function $s(\cdot,\cdot)$ (e.g., cosine similarity) measures the alignment between these embeddings, and τ_m is a temperature parameter.

Overlapping users are used in two components of PACL-CDR: (1) the cross-domain item similarity $\mathrm{Sim}(v_S, v_T)$ in the tag construction of DPM (Eq. (17)), and (2) the positive pairs in L_{II} (Eq. (28)). Beyond these anchor points, cross-domain knowledge is transferred through intent-level representations

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Algorithm 1 Training Procedure of PACL-CDR (Summarized)

Require: Source-domain data $\mathcal{D}^S$, target-domain data $\mathcal{D}^T$, cross-domain training loader $\mathcal{D}^{train}$, graphs $\mathcal{G}^S$ and $\mathcal{G}^T$, and model parameters Θ.
Ensure: Optimized model parameters Θ.

```
Randomly initialize all learnable parameters Θ.
for epoch = 1, 2, ..., E do
    for each mini-batch B sampled from D^train do
        Reset the optimizer gradients and initialize L_total = 0.
        // Step 1: Domain-specific representation learning
        for each domain d ∈ {S, T} do
            Obtain user and item embeddings from graph G^d using the GNN encoder.
            Compute graph denoising contrastive loss L^d_DCL.
            // Step 2: Debiasing Popularity Module
            Compute popularity-adaptive losses L^d_SAS + L^d_RCS.
            Construct item tag representations based on the popularity-debiased item embeddings.
            Compute tag-level contrastive loss L^d_TE.
            // Step 3: Intent Enhancement Module
            Construct intent-level user representations and store them as z_d.
            Compute intent-level contrastive loss L^d_IE.
            Compute BPR-based recommendation loss L^d_rec.
            Accumulate domain-specific losses: L_total ← L_total + L^d_rec + λ1 L^d_DCL + (L^d_SAS + L^d_RCS) + λ2(L^d_IE + L^d_TE).
        end for
        // Step 4: Cross-domain intent alignment
        Compute cross-domain intent alignment loss L_II between Int^S_{u_i} and Int^T_{u_i}.
        Accumulate: L_total ← L_total + L_II.
        // Step 5: Parameter update
        Backpropagate L_total and update Θ.
    end for
end for
return Optimized parameters Θ.
```

rather than additional direct user-to-user alignment. PACL-CDR therefore remains dependent on user overlap, and its robustness to limited cross-domain correspondence is evaluated empirically in Section III.E.

The expected benefit of IEM arises from transferring knowledge at the intent level. The association scores γ_u map each user's behavior to global latent intent prototypes, while L_{IE} aligns the global user embedding with the corresponding aggregated intent representation and distinguishes it from those of other users. Based on overlapping users, L_{II} further aligns intent representations across domains. Consequently, the transferred information is a structured intent representation rather than a direct alignment of unconstrained ID-based user embeddings. This design provides auxiliary supervision for users with sparse interactions and may reduce sensitivity to individual overlap correspondences, offering a representation-level explanation for improved performance under limited cross-domain correspondence without implying a formal generalization guarantee.

TABLE I: Statistics of Cross-Domain Recommendation Tasks for Amazon Datasets

Tasks	Dataset	$\lvert\mathcal{U}\rvert$	$\lvert\mathcal{V}\rvert$	Training	Test	Density
Task 1	Electronic	3,325	17,709	50,407	2,559	0.089%
	Phone	3,325	38,706	115,554	2,560	0.091%
Task 2	Sport	9,928	30,796	92,612	8,326	0.033%
	Cloth	9,928	39,008	87,829	7,540	0.024%
Task 3	Sport	4,998	20,845	50,558	3,698	0.052%
	Phone	4,998	13,655	42,446	3,999	0.068%
Task 4	Electronic	15,761	51,447	210,865	13,824	0.052%
	Cloth	15,761	48,781	121,083	12,526	0.068%

Finally, we jointly optimize the main recommendation objective and the auxiliary contrastive and regularization objectives, as summarized in Algorithm 1. The primary recommendation objective adopts the Bayesian Personalized Ranking (BPR) loss $\mathcal{L}_{\text{rec}}$, which encourages interacted items to receive higher predicted scores than unobserved items. The overall training objective is defined as

$$L_{\text{total}} = \sum_{d\in\{S,T\}} \Big[L^d_{\text{rec}} + \lambda_1 L^d_{\text{DCL}} + L^d_{\text{SAS}} + L^d_{\text{RCS}} + \lambda_2 \big(L^d_{\text{IE}} + L^d_{\text{TE}} \big) \Big] + L_{\text{II}}, \tag{29}$$

where d denotes the source (S) or target (T) domain. The hyperparameters λ_1 and λ_2 balance the contributions of the corresponding auxiliary objectives. L_{SAS} and L_{RCS} are incorporated without additional coefficients and serve as intrinsic regularizers for popularity-aware representation learning, while L_{II} directly constrains cross-domain intent alignment. Joint optimization enables the recommendation, popularity-aware, and intent-aware objectives to collaboratively guide model training.

E. Computational Complexity Analysis

Let $N = |\mathcal{U}|+|\mathcal{V}|$ denote the total number of users and items across the two domains, $|\mathcal{E}|$ the number of observed interactions, L the number of GNN layers, d the embedding dimension, K the number of latent prototypes, B the mini-batch size, and M the number of mini-batches per epoch. The graph encoder requires $\mathcal{O}\big(L(|\mathcal{E}|d + Nd^2)\big)$ operations for graph propagation. The intent- and tag-projection layers introduce an additional cost of $\mathcal{O}(MBKd^2)$, while the in-batch contrastive objectives require $\mathcal{O}(MB^2d)$. The resulting training complexity is

$$\mathcal{O}\big(L(|\mathcal{E}|d + Nd^2) + M(BKd^2 + B^2d)\big). \tag{30}$$

The additional parameter complexity is $\mathcal{O}(Kd^2)$, and the principal intermediate-memory overhead is $\mathcal{O}(BKd + B^2)$. These additional computations are primarily incurred during training; once the representations are learned, the auxiliary contrastive objectives are not evaluated during ranking, and user–item scoring retains a cost of $\mathcal{O}(d)$ per pair.

III. Experiments

To demonstrate the effectiveness of the proposed PACL-CDR model, we design our experiments to address the following research questions:

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TABLE II: Performance comparison of different methods across multiple datasets. Metrics include HR@10 and NDCG@10. The best results are in **bold**, and the second-best results are underlined.

Task	**Task1**				**Task2**				**Task3**				**Task4**			
Dataset	Electronic		Phone		Sport		Cloth		Sport		Phone		Electronic		Cloth	
Metric@10	HR	NDCG	HR	NDCG	HR	NDCG	HR	NDCG	HR	NDCG	HR	NDCG	HR	NDCG	HR	NDCG
NeuMF	16.17	9.24	15.84	8.02	10.74	5.56	11.18	6.02	10.11	5.19	14.67	7.80	20.08	11.79	10.84	5.80
NGCF	18.55	10.87	22.79	12.38	13.13	6.87	13.22	6.97	16.06	8.53	17.07	9.22	20.20	11.74	10.86	5.80
LightGCN	19.71	10.28	23.25	12.72	13.19	6.94	13.58	7.29	16.33	9.16	16.47	8.95	19.97	10.73	11.24	6.11
DGCF	19.69	11.69	24.89	14.88	15.67	7.74	12.27	6.82	18.82	11.47	19.90	11.82	16.45	9.98	9.24	5.55
DICE	10.12	6.19	22.00	13.09	14.63	7.85	12.27	6.65	15.31	9.01	19.80	11.31	18.14	11.45	9.80	5.54
CoNet	17.22	9.86	17.66	9.30	15.05	7.42	10.23	4.47	12.88	6.91	16.60	9.15	21.26	12.61	11.35	6.19
DDTCDR	18.47	11.08	17.23	8.58	11.86	6.37	12.54	7.13	12.14	6.47	16.17	8.98	21.70	13.10	11.47	6.38
PPGN	21.68	11.63	24.54	13.34	12.03	6.42	10.52	4.78	18.00	10.54	20.40	11.09	21.85	12.36	12.98	6.88
BiTGCF	22.14	12.52	25.17	13.93	12.11	6.46	10.58	4.93	18.63	10.11	21.10	11.25	21.61	12.25	13.11	6.80
DisenCDR	24.54	13.23	26.77	14.95	16.39	8.70	14.56	8.20	19.66	11.09	21.96	11.80	22.67	13.14	13.60	7.37
DIDA-CDR	21.21	12.52	25.17	13.94	18.15	9.49	13.73	7.63	18.90	10.80	20.21	11.48	22.53	13.36	13.42	7.36
UniCDR	21.75	13.29	23.95	14.03	21.87	11.91	17.11	10.19	19.33	11.27	21.23	12.20	23.84	14.06	12.42	6.94
PPA	25.07	14.63	28.56	16.04	21.01	12.39	20.07	12.32	17.84	10.27	21.53	11.95	20.23	11.20	9.95	5.63
CVGAE	21.13	12.30	30.05	17.26	22.05	11.65	19.83	10.73	18.63	11.88	22.07	12.31	19.77	10.44	13.32	**8.96**
PACL-CDR	**26.10**	**16.22**	**31.43**	**18.76**	**23.49**	**14.47**	**21.52**	**13.53**	**20.66**	**12.58**	**24.20**	**13.86**	**24.29**	**14.70**	**14.92**	8.59
± **SD**	**±0.09**	**±0.07**	**±0.09**	**±0.06**	**±0.13**	**±0.07**	**±0.11**	**±0.12**	**±0.14**	**±0.13**	**±0.10**	**±0.03**	**±0.15**	**±0.05**	**±0.20**	**±0.11**
Imp. ↑	4.11%	10.87%	4.59%	8.69%	6.53%	16.79%	7.22%	9.82%	5.09%	5.89%	9.65%	12.59%	1.89%	4.55%	9.71%	–

- **RQ1**: How does PACL-CDR compare with representative single-domain and cross-domain baselines in overall ranking performance across the four evaluated tasks? (Section III-B)
- **RQ2**: What are the individual and complementary contributions of denoising, DPM, and IEM to ranking performance? (Section III-C)
- **RQ3**: How does DPM affect head–tail ranking quality and tail-item exposure? (Section III-D)
- **RQ4**: How robust is PACL-CDR under interaction noise, within-domain interaction sparsity, and limited cross-domain user correspondence? (Section III-E)
- **RQ5**: What training and inference overhead does PACL-CDR introduce compared with representative CDR baselines? (Section III-F)
- **RQ6**: How sensitive is the ranking performance of PACL-CDR to its key hyperparameters? (Section III-G)
- **RQ7**: What qualitative structure is observed in the two-dimensional projections of the learned user-intent and item-tag embeddings? (Section III-H)

A. Experimental Settings

*1) **Tasks and Datasets**:* We construct four CDR tasks from the Amazon dataset[1] following DisenCDR [24]: **Electronic&Phone, Sport&Cloth, Sport&Phone, and Electronic&Cloth**. For each domain, we independently retain users with at least 5 interactions and items with at least 10 interactions, and then extract the overlapping users between the paired domains. Since the filtering is performed separately within each domain, all evaluated users have historical interactions in both domains; therefore, the setting does not include strict zero-interaction cold-start users. Table I reports the dataset statistics.

[1]http://jmcauley.ucsd.edu/data/amazon/index_2014.html

*2) **Evaluation Metrics**:* Following prior studies, we evaluate ranking performance using HR and NDCG. HR measures whether the ground-truth item appears in the recommendation list, while NDCG additionally considers its ranking position. For each validation and test instance, we rank one positive item together with 999 randomly sampled negative items and report HR@10 and NDCG@10.

To evaluate the exposure of low-popularity items, we additionally adopt Fair-α@K [25]. Given the Top-K recommendation list R_u^K for each warm user $u \in \mathcal{U}_{\text{warm}}$, it is defined as

$$\text{Fair-}\alpha@K = \frac{\sum_{u\in\mathcal{U}_{\text{warm}}} \sum_{i\in R_u^K} \mathbb{I}\big(i \in \mathcal{I}_{\text{tail}}^{\text{train}}(\alpha)\big)}{\sum_{u\in\mathcal{U}_{\text{warm}}} |R_u^K|}. \quad (31)$$

Here, $\mathcal{I}_{\text{tail}}^{\text{train}}(\alpha)$ denotes the fixed bottom-α proportion of training-observed items ranked by interaction frequency. The metric therefore represents the proportion of Top-10 recommendation positions occupied by items in the training-derived bottom-80% tail set under the same sampled-candidate protocol. It evaluates popularity-related item exposure rather than demographic or user-group fairness.

*3) **Comparison with State-of-the-Art Methods**: **Single-domain baselines:*** We compare PACL-CDR with five representative single-domain recommendation models. **NeuMF** [26] combines generalized matrix factorization and multi-layer perceptrons to model non-linear user–item interactions. **NGCF** [27] employs graph neural networks to capture high-order connectivity in user–item interaction graphs. **LightGCN** [28] simplifies graph convolution by retaining only linear neighborhood aggregation and removing feature transformations and non-linear activations. **DGCF** [29] constructs intent-aware interaction graphs and encourages independence among latent intents to learn disentangled representations. **DICE**

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TABLE III: Performance comparison and ablation study results (HR@10 and NDCG@10). Four ablation settings are considered: (1) **w/o De-noise**: removing the denoising module; (2) **w/o DPM**: removing the Debiasing Popularity Module; (3) **w/o IEM**: removing the Intent Enhancement Module; and (4) **w/o DPM & IEM**: removing both the Debiasing Popularity Module and the Intent Enhancement Module. The percentage drops (%) are computed relative to the full model (PACL-CDR).

Task	Dataset	Metric@10	PACL-CDR	w/o De-noise		w/o DPM		w/o IEM		w/o DPM & IEM	
				Value	Drop (%)↓	Value	Drop (%)↓	Value	Drop (%)↓	Value	Drop (%)↓
Task 1	Electronic	HR	26.10 ± 0.09	24.81 ± 0.18	4.94	24.93 ± 0.18	4.48	25.29 ± 0.10	3.10	24.62 ± 0.28	5.67
		NDCG	16.22 ± 0.09	15.62 ± 0.14	3.70	15.56 ± 0.10	4.07	15.38 ± 0.12	5.18	15.27 ± 0.17	5.86
	Phone	HR	31.43 ± 0.09	31.27 ± 0.10	0.51	30.78 ± 0.28	2.07	30.26 ± 0.17	3.72	30.06 ± 0.17	4.36
		NDCG	18.76 ± 0.06	18.39 ± 0.11	1.97	18.15 ± 0.24	3.25	18.15 ± 0.17	3.25	17.71 ± 0.16	5.60
Task 2	Sport	HR	23.49 ± 0.13	23.27 ± 0.30	0.94	23.27 ± 0.12	0.94	23.31 ± 0.20	0.77	21.71 ± 0.07	7.58
		NDCG	14.47 ± 0.07	14.29 ± 0.08	1.24	14.18 ± 0.08	2.00	14.28 ± 0.04	1.31	13.30 ± 0.15	8.09
	Cloth	HR	21.52 ± 0.11	21.42 ± 0.12	0.46	21.27 ± 0.11	1.16	21.10 ± 0.07	1.95	20.67 ± 0.15	3.95
		NDCG	13.53 ± 0.12	13.23 ± 0.04	2.22	13.30 ± 0.05	1.70	13.24 ± 0.06	2.14	12.84 ± 0.16	5.10
Task 3	Sport	HR	20.66 ± 0.14	20.19 ± 0.13	2.28	20.36 ± 0.24	1.45	20.19 ± 0.14	2.28	19.88 ± 0.13	3.77
		NDCG	12.58 ± 0.13	12.23 ± 0.15	2.78	12.29 ± 0.18	2.31	12.38 ± 0.10	1.59	11.84 ± 0.18	5.88
	Phone	HR	24.20 ± 0.10	23.68 ± 0.27	2.15	23.18 ± 0.07	4.21	23.86 ± 0.10	1.40	23.37 ± 0.21	3.43
		NDCG	13.86 ± 0.03	13.67 ± 0.10	1.37	13.51 ± 0.14	2.53	13.47 ± 0.08	2.81	12.91 ± 0.05	6.85
Task 4	Electronic	HR	24.29 ± 0.15	23.58 ± 0.11	2.92	23.18 ± 0.07	4.57	23.85 ± 0.10	1.81	23.51 ± 0.07	3.21
		NDCG	14.70 ± 0.05	14.42 ± 0.09	1.90	14.58 ± 0.09	0.82	14.34 ± 0.06	2.45	14.37 ± 0.09	2.24
	Cloth	HR	14.92 ± 0.20	14.51 ± 0.10	2.75	14.42 ± 0.08	3.35	14.15 ± 0.03	5.16	13.81 ± 0.10	7.44
		NDCG	8.59 ± 0.11	8.46 ± 0.10	1.51	8.21 ± 0.04	4.42	8.40 ± 0.08	2.21	8.09 ± 0.10	5.82

[30] separates user interest from conformity and learns the corresponding representations using cause-specific data derived from causal inference.

Cross-domain baselines: We further compare PACL-CDR with nine representative cross-domain recommendation methods. **CoNet** [31] uses a customized cross-stitch architecture with deep cross-connections to transfer information across domains. **DDTCDR** [32] introduces a deep dual transfer network that mutually reinforces knowledge learning in the source and target domains. **PPGN** [33] combines a Cross-Domain Preference Matrix with a propagation layer to model and transfer user preferences. **BiTGCF** [34] captures high-order connectivity within domain-specific graphs and performs bidirectional knowledge transfer through shared users. **DisenCDR** [24] uses a mutual-information-based objective to disentangle domain-shared and domain-specific user representations. **DIDA-CDR** [32] applies interpolative data augmentation to generate user representations and separate domain-specific from domain-independent information. **UniCDR** [35] integrates domain-specific and domain-shared features within a unified framework for cross-domain knowledge transfer. **PPA** [36] learns high-level user preference representations from the source domain and transfers them to the target domain. **CVGAE** [1] combines variational autoencoders and graph convolutional networks for cross-domain representation learning while protecting user attributes through a camouflaging mechanism.

4) Implementation Details: We implement PACL-CDR in PyTorch on an NVIDIA RTX 4090 GPU, trained for 100 epochs with the Adam optimizer (batch size 1024, learning rate 1×10^{-3}). Embedding and hidden dimensions are set to 128. Key hyperparameters, tuned via grid search, are: momentum $\alpha = 0.7$, interpolation $\beta = 0.7$, neighborhood size $k = 15$, 15 latent intents, mask temperature $\tau = 0.45$, contrastive temperature $\tau_m = 0.4$, and loss coefficients $\lambda_1 = 0.6$, $\lambda_2 = 0.8$. Data is split 7:1:2 for training, validation, and testing; results are averaged over five independent runs.

B. Experimental Results and Analysis (RQ1)

RQ1 evaluates whether PACL-CDR improves overall ranking performance compared with single-domain and cross-domain baselines across four Amazon cross-domain recommendation tasks. Table II reports HR@10 and NDCG@10, with the PACL-CDR results averaged over five independent runs. The accompanying standard deviations summarize run-to-run variability across the five independent runs.

1) Single-Domain Results and Analysis: PACL-CDR consistently outperforms the single-domain baselines under the evaluated settings. For example, in Task 2, it achieves HR@10 and NDCG@10 scores of 23.49 and 14.47 on Sport, and 21.52 and 13.53 on Cloth, respectively, exceeding NeuMF and LightGCN. These results indicate that information from a related domain can supplement sparse within-domain interactions and improve ranking performance.

2) Cross-Domain Results and Analysis: Compared with cross-domain baselines such as DisenCDR, UniCDR, PPA, and CVGAE, PACL-CDR achieves the best mean result on most reported metrics. In Task 3, it obtains HR@10 scores of 20.66 on Sport and 24.20 on Phone, exceeding the respective second-best results by 5.09% and 9.65%. These results indicate the competitive overall ranking performance of the integrated framework under the evaluated settings. PACL-CDR does not achieve the best performance in every setting. Its NDCG10 score of 8.59 on Cloth in Task 4 is slightly lower than the 8.96 achieved by CVGAE. Nevertheless, its overall results across the four tasks confirm its competitive ranking performance.

C. Ablation Study Analysis (RQ2)

RQ2 examines the contribution of each component of PACL-CDR under the same experimental settings. We construct four variants: w/o De-noise removes the graph-denoising mechanism and L_{DCL}; w/o DPM removes the popularity-aware objectives L_{SAS}, L_{RCS}, and L_{TE}; w/o IEM removes intent representation

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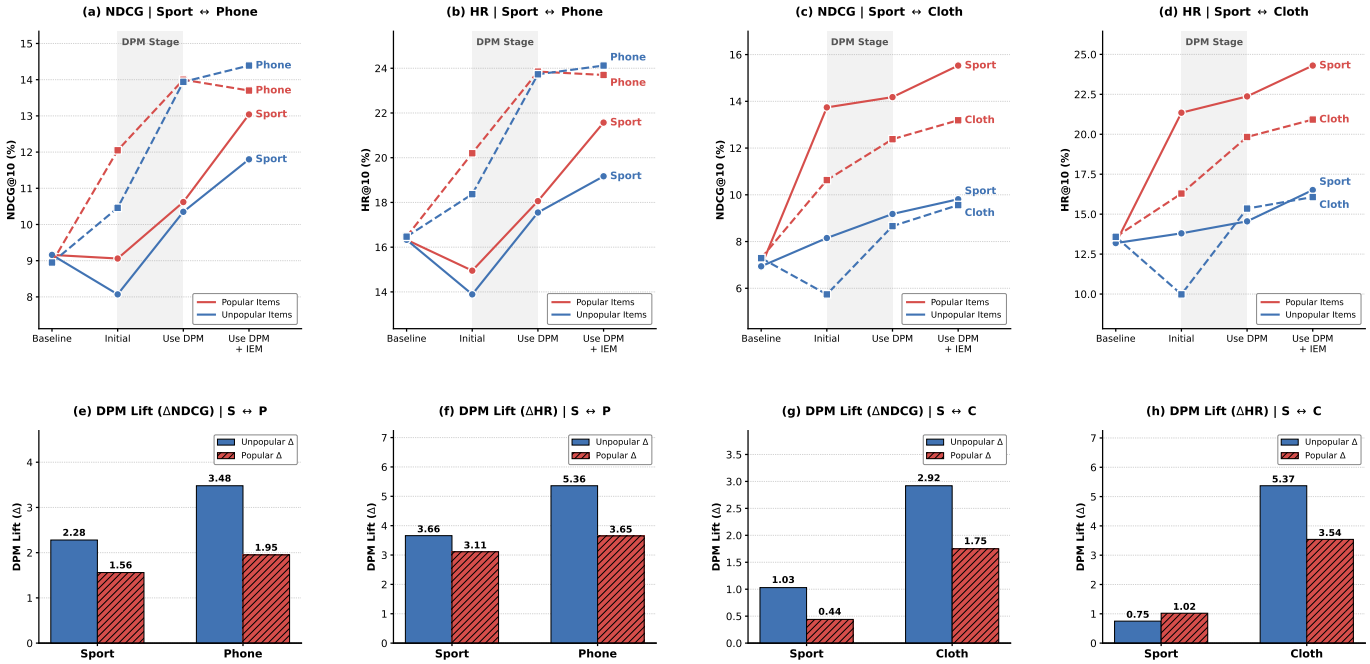


Fig. 3: Head–tail stratified ranking results in the Sport↔Phone and Sport↔Cloth scenarios. Panels (a)–(d) compare Baseline (vanilla GNN), Initial (pre-DPM), +DPM, and +DPM&IEM, while panels (e)–(h) report the absolute DPM lift (Δ = +DPM − Initial) for the two item groups.

TABLE IV: Fair-0.8@10 (%) on the four cross-domains. Results are reported as mean ± standard deviation over five independent runs; higher values indicate greater exposure of the fixed training-derived tail set.

Model	Task 1		Task 2		Task 3		Task 4	
	Electronic	Phone	Sport	Cloth	Sport	Phone	Electronic	Cloth
DisenCDR	1.2526 ± 0.2319	1.3286 ± 0.3492	2.4061 ± 0.0988	4.1561 ± 0.6311	5.0469 ± 0.8063	2.6590 ± 0.7444	5.1479 ± 0.0002	7.3080 ± 0.1986
UniCDR	2.0677 ± 0.7103	5.6246 ± 0.6070	11.7456 ± 0.1090	9.4607 ± 0.3931	10.4615 ± 0.8331	5.2730 ± 1.2493	5.8529 ± 0.0382	9.4560 ± 0.3366
PPA	3.8906 ± 0.0959	4.8990 ± 0.1179	11.0137 ± 0.2314	9.2109 ± 0.1264	8.4469 ± 0.1913	4.7112 ± 0.0524	4.2175 ± 0.0679	9.2118 ± 0.1329
CVGAE	3.0886 ± 0.0825	4.7315 ± 0.2577	10.2411 ± 0.4026	9.6472 ± 0.2859	7.3263 ± 0.2253	5.7719 ± 0.9929	7.1963 ± 0.0579	$\mathbf{12.4694 \pm 0.4752}$
PACL-CDR w/o DPM	4.3724 ± 0.1579	6.2054 ± 0.2686	11.5639 ± 0.4704	10.2162 ± 0.0790	10.2957 ± 0.2806	5.8273 ± 0.0472	8.0618 ± 0.1412	12.2487 ± 0.1287
PACL-CDR	$\mathbf{4.6849 \pm 0.1760}$	$\mathbf{6.8464 \pm 0.1241}$	$\mathbf{12.2380 \pm 0.1573}$	$\mathbf{10.5314 \pm 0.0628}$	$\mathbf{10.7680 \pm 0.2719}$	$\mathbf{6.0882 \pm 0.3540}$	$\mathbf{8.3722 \pm 0.1763}$	12.3852 ± 0.1740

learning and cross-domain intent alignment through L_{IE} and L_{II}; and w/o DPM & IEM removes both modules. Table III reports the relative decreases in HR@10 and NDCG@10 compared with the full model.

Removing any component decreases both metrics across the evaluated tasks, although the magnitude differs among settings. On Phone in Task 1, removing the denoising mechanism reduces HR@10 and NDCG@10 by 0.51% and 1.97%, respectively. Removing DPM leads to decreases of 2.07% and 3.25%, while removing IEM results in decreases of 3.72% and 3.25%. These results indicate that denoising, popularity-aware item learning, and intent-level user modeling each contribute to ranking performance. The largest degradation occurs when DPM and IEM are removed together, with HR@10 and NDCG@10 decreasing by 7.58% and 8.09% on Sport in Task 2, demonstrating their complementary contributions to PACL-CDR.

D. Popularity Bias Analysis (RQ3)

RQ3 evaluates DPM from two complementary perspectives. Fig. 3 uses staged ablations (Baseline, Initial, +DPM, and +DPM&IEM) to measure HR@10 and NDCG@10 separately for head and tail items, thereby examining ranking quality within each popularity group. Table IV reports Fair-0.8@10, which measures the proportion of Top-10 positions assigned to the fixed training-derived tail set. Thus, the figure evaluates group-specific ranking quality, whereas the table evaluates tail-item exposure.

Relative to the Initial stage, adding DPM improves tail-item NDCG@10 and HR@10 by 2.28 and 3.66 points on Sport and by 3.48 and 5.36 points on Phone in the Sport↔Phone scenario. In the Sport↔Cloth scenario, adding DPM improves tail-item NDCG@10 and HR@10 by 1.03 and 0.75 points on Sport and by 2.92 and 5.37 points on Cloth, respectively. Thus, the tail-item ranking effects are positive across all evaluated domain–metric combinations in the two illustrated scenarios.

Table IV shows that PACL-CDR achieves higher mean Fair-0.8@10 than PACL-CDR w/o DPM in all eight source- and target-domain evaluations, with gains of 0.1365–0.6741 percentage points. This comparison associates DPM with increased exposure of training-observed tail items. Together, the figure and table show that DPM improves tail-item ranking

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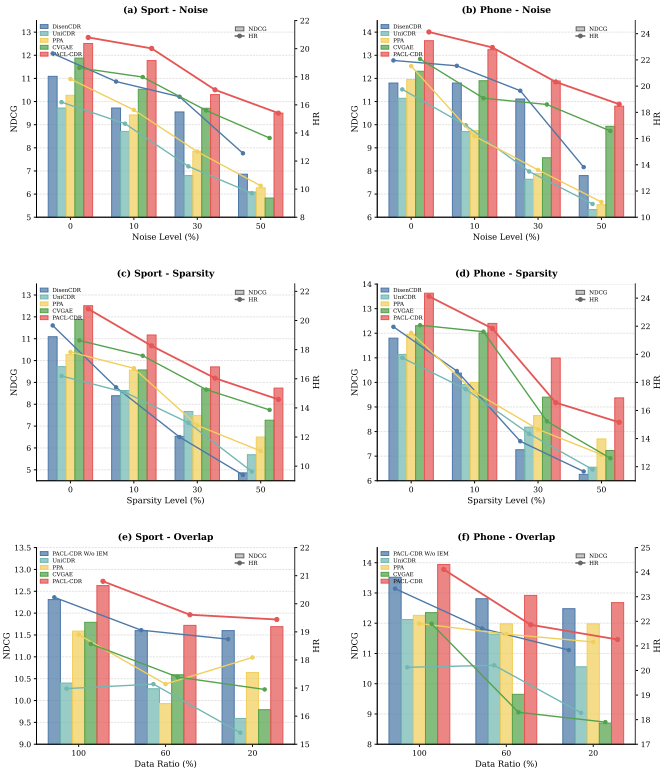


Fig. 4: Robustness evaluation in the Sport↔Phone scenario. Panels (a)–(d) examine interaction noise and within-domain sparsity, while panels (e)–(f) evaluate retained overlapping-user links. PACL-CDR w/o IEM isolates the effect of intent enhancement.

quality in the two illustrated scenarios and is associated with higher mean tail exposure in all eight evaluations under the sampled-candidate protocol, without implying that popularity bias is eliminated.

E. Robustness Evaluation (RQ4)

To assess robustness under distinct sources of missing or corrupted information, we evaluate PACL-CDR under interaction noise, within-domain interaction sparsity, and controlled removal of cross-domain user links. The third setting differs from general sparsity because it masks user identity correspondences across domains while retaining the associated within-domain interaction histories. Fig. 4 reports the results under these three conditions.

(1) Performance Under Noisy Data. We conduct noise-injection experiments by randomly corrupting 10%, 30%, and 50% of the interactions. PACL-CDR maintains strong performance in both evaluated domains. At the 50% noise level, it achieves an NDCG of 9.50 and HR of 15.41 in Sport and an NDCG of 10.80 and HR of 18.62 in Phone, exceeding the compared baselines in this setting.

(2) Performance Under Sparse Data. By constructing sparse datasets that retain only 10%, 30%, and 50% of the original interactions, we observe a smaller performance degradation for PACL-CDR than for the compared methods. For instance, at the reported 50% sparsity setting in Sport, NDCG decreases from 12.51 to 8.74 (a drop of 3.77) while remaining the highest among the compared results. The corresponding HR trends provide additional evidence of robustness under the evaluated sparsity levels.

TABLE V: Computational-efficiency comparison under the common hardware and experimental protocol. Training and inference times are measured in minutes, and peak GPU memory is measured during training.

Model	Parameters	Time/Epoch	Peak Memory (MiB)	Inference Time
DisenCDR	14.58M	0.56	1,444	0.09
UniCDR	12.15M	0.56	774	0.14
PPA	16.47M	1.52	3,874	0.14
CVGAE	23.17M	1.40	1,048	0.04
PACL-CDR	41.05M	3.01	2,762	0.13

(3) Performance Under Severed Cross-Domain User Links. To isolate missing cross-domain user correspondence from general interaction sparsity, we conduct a controlled link-severing experiment with retention ratios $q \in \{100\%, 60\%, 20\%\}$. At each ratio, only the fraction q of the original overlapping-user identity links remains available as cross-domain training anchors. For each severed link, the domain-specific user records and all within-domain interactions are retained, but the mapping that identifies the two records as belonging to the same user is hidden during training. Thus, at 20% retention, 80% of the original identity mappings are unavailable, while neither the users nor their within-domain interactions are removed. This design varies only the availability of cross-domain user correspondence.

As shown in Fig. 4(e)–(f), PACL-CDR outperforms PACL-CDR w/o IEM in HR@10 and NDCG@10 on both Sport and Phone at all three retention ratios. The full model also remains the best-performing method when only 20% of the original identity links are retained. Because PACL-CDR and its w/o IEM variant are evaluated under the same link-retention settings and differ only in the intent enhancement component, their comparison provides controlled ablation evidence that IEM contributes to robustness when cross-domain user correspondences are limited. This result supports reduced sensitivity to missing identity links, rather than complete independence from overlapping users.

Overall, PACL-CDR maintains competitive recommendation performance under interaction noise, within-domain interaction sparsity, and limited cross-domain user correspondence.

F. Computational Efficiency (RQ5)

As shown in Table V, PACL-CDR contains 41.05M parameters and requires 3.01 minutes per training epoch, reflecting the additional projection layers and contrastive objectives introduced by DPM and IEM. Its peak training memory is 2,762 MiB, which is 28.7% lower than PPA's 3,874 MiB but higher than those of DisenCDR, UniCDR, and CVGAE. PACL-CDR requires 0.13 minutes for inference, close to UniCDR and PPA at 0.14 minutes, although slower than DisenCDR and CVGAE. These results indicate that the principal computational trade-off of PACL-CDR lies in its parameter count and training time, whereas its measured inference cost remains within the same order of magnitude as the representative baselines.

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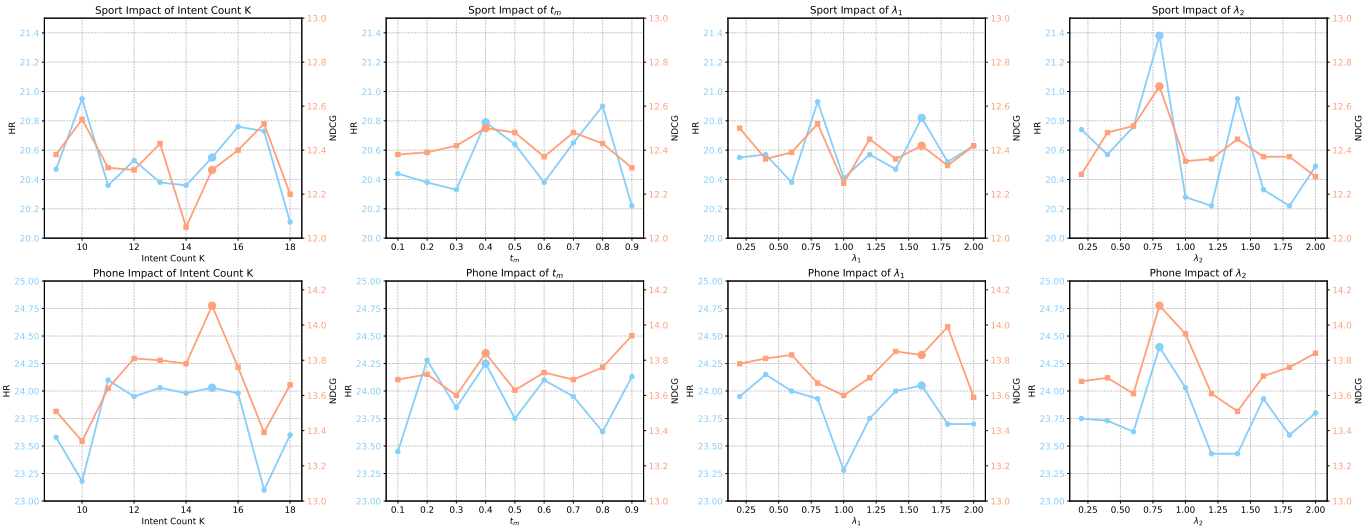


Fig. 5: Performance and runtime comparison with different parameter settings for the Sport&Phone scenario.

All methods are trained under the common 100-epoch protocol. Consequently, the reported time per epoch characterizes computational overhead but should not be interpreted as a comparison of convergence speed.

G. Parameter Sensitivity (RQ6)

We select hyperparameters globally rather than tuning them separately for each domain. This protocol reduces the risk of domain-specific overfitting and tests whether a common configuration can maintain competitive performance across the evaluated domains; it does not constitute a general guarantee of cross-domain adaptability.

Number of intents K. This parameter controls the model's capacity to capture diverse user interests (see Fig. 5). Setting $K = 15$ achieves an optimal balance, yielding an HR of 20.55 in the Sport domain and 23.10 in the Phone domain. Too few intents fail to capture fine-grained preferences, while excessively many introduce redundant representations, increasing complexity and hurting generalization.

Temperature parameter τ_m. This regulates the sensitivity of contrastive learning. We set $\tau_m = 0.4$ to achieve the best cross-domain trade-off (HR of 24.25 on Phone and 20.79 on Sport). Overly small values overemphasize hard negatives and weaken generalization, whereas excessively large temperatures blur the distinction between positive and negative samples, making representation learning difficult.

Denoising strength λ_1. This coefficient controls the suppression of irrelevant or noisy features. Setting $\lambda_1 = 0.6$ strikes the best balance across the Sport and Phone domains. Smaller values fail to suppress noise adequately, while larger values risk over-filtering useful signals and causing underfitting.

Intent modeling weight λ_2. This determines the contribution of the intent modeling loss. $\lambda_2 = 0.8$ yields the optimal HR performance (21.38 on Sport and 24.40 on Phone). Smaller values weaken the model's ability to capture user interests, whereas excessively large values overshadow and impair the effectiveness of other components.

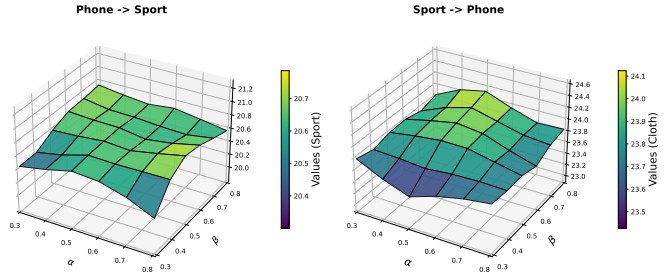


Fig. 6: Study on the weight parameters α and β in terms of the NDCG@10 metric.

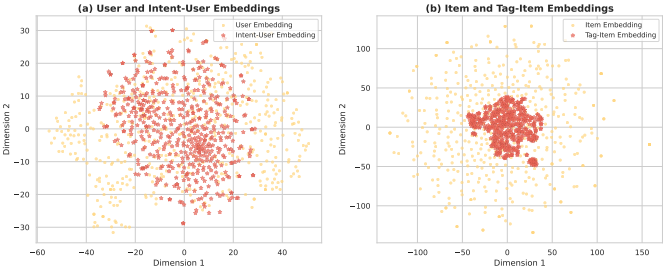


Fig. 7: t-SNE visualization of user-item embeddings in the sport&phone scenario.

Weight distribution parameters α and β. These parameters control the balance between popular and unpopular samples within contrastive pairs (see Fig. 6). Setting $\alpha = \beta = 0.7$ optimally balances head and tail items (HR of 21.11 on Sport and 24.58 on Cloth). Smaller values over-penalize popular samples, while larger values neglect unpopular items, thereby reducing recommendation diversity.

H. Visualization Analysis (RQ7)

Fig. 7 illustrates changes in user and item embedding distributions after intent and tag modeling. Subfigure (a) compares initial user embeddings with intent-user embeddings,

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while Subfigure (b) compares item embeddings with tag-item embeddings. In both cases, the latter embeddings form more compact clusters in the displayed two-dimensional projection. This visualization provides qualitative evidence of representation restructuring, but it is not treated as proof of improved generalization or recommendation performance.

IV. Related Works

A. Cross-Domain Recommendation

Cross-domain recommendation (CDR) [4], [5] addresses data sparsity and cold-start challenges by transferring knowledge from data-rich source domains to data-sparse target domains. Early research focused on leveraging overlapping users or items to align user preferences across domains. For instance, DA-CDR [2] improves cross-domain recommendations by using dual adversarial learning to align shared user–item representations while extracting domain-specific features for bidirectional knowledge transfer. Similarly, CDLFM [37] improvs recommendations for new users by aligning user representations via latent feature mapping. More recently, CUT [38] introduced a framework that reduces negative transfer by filtering irrelevant collaborative information from the source domain. This framework uses target-domain user similarity as a constraint and incorporates a user transformation layer to guide cross-domain information transfer, thereby improving performance in multi-domain recommendation tasks.

When addressing cold-start problems, noise interference during knowledge transfer and mismatches in extracted preference features remain significant challenges. Methods like MUCRP [39] aligns global and domain-specific preferences using fused Gromov-Wasserstein optimal transport, unifying user behaviors in a shared latent space. RPT [40] employs a multi-task self-supervised pre-training framework that integrates user and community data using hierarchical Transformers and graph encoders. By using contrastive learning as its core objective and auxiliary tasks to enhance training, RPT effectively transfers to downstream tasks, significantly improving performance in scholarly data mining tasks.

To address negative transfer, models like TrineCDR [41] systematically identify its underlying causes and improve knowledge transferability across the feature, interaction, and domain levels. Meanwhile, UniCDR [35] combines multi-domain interactions to extract shared user behaviors. Despite these advancements, significant challenges remain, particularly in achieving effective cross-domain alignment and reducing noise interference.

Contrastive Cross-Domain Recommendation (CL-CDR) applies contrastive learning techniques to address challenges in semantic alignment and negative transfer in cross-domain recommendation. Methods like RecDCL [42] and DisCo [12] employ dual contrastive loss and graph-based disentangled contrastive learning frameworks, respectively, to capture fine-grained user intents and align behaviors across domains. These approaches enhance cross-domain consistency but often overemphasize local preferences, neglecting global relationships.

Representative CDR methods differ substantially in their methodological objectives. DisenCDR [24] disentangles domain-shared and domain-specific information, UniCDR [35] unifies data-sparsity and cold-start scenarios, PPA [36] transfers quantified preference prototypes, and CVGAE [1] protects user attributes during graph-based transfer. In contrast, PACL-CDR explicitly introduces item popularity into supervised alignment and contrastive reweighting, while IEM represents user behaviors through global latent intent prototypes and aligns intent representations across domains. Hence, PACL-CDR is distinguished by its popularity-adaptive optimization and the complementary division between item-side debiasing and intent-level transfer, rather than by the simple combination of existing components.

B. Popularity Bias in Recommendation

Popularity Bias [19]–[21] is a critical challenge in recommender systems, where highly popular items dominate recommendation results, leaving unpopular items significantly underrepresented. This bias stems from the skewed distribution of user interactions in the training dataset, where popular items naturally receive more feedback, leading algorithms to prioritize them during learning. The issue is particularly problematic in domains where users' interests are diverse, as it limits the discovery of niche or long-tail items that could better match user preferences. Various techniques have been proposed to analyze and mitigate popularity bias, broadly categorized into three types:

Re-weighting-based methods increase the training weights or scores of unpopular items to reduce the focus on popular items. For instance, IPS [22] compensates for unpopular items and adjusts the user-item preference matrix, improving their rankings in the recommendation results. Decorrelation-based methods aim to eliminate the correlation between item representations and popularity. For example, MACR [43] employs counterfactual reasoning to remove the direct impact of popularity on recommendation outcomes, ensuring unbiased item representations. Contrastive learning-based methods enhance the uniformity of item representations to alleviate popularity bias. For instance, SimGCL [44] utilizes InfoNCE [45] loss with data augmentation techniques to generate diverse views, promoting consistent representations for unpopular items while mitigating the influence of popularity.

However, these works mainly focus on single-domain recommendation. When transferring knowledge from a source domain to a target domain, popularity bias can be further amplified across domains, which has been largely overlooked. Our work is among the first to explicitly address cross-domain popularity bias under a contrastive learning framework.

V. Conclusion and Future Work

This paper introduces PACL-CDR, a cross-domain recommendation framework that combines popularity-adaptive item optimization with intent-level user transfer. Across four datasets, PACL-CDR achieves the strongest mean ranking performance in most evaluated settings. The Fair-0.8@10 results and the w/o DPM comparison associate DPM with increased exposure of the fixed training-derived tail set, while the controlled link-severing experiment and the w/o IEM comparison show

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that IEM contributes to reduced sensitivity to limited cross-domain user correspondence. These findings are restricted to the evaluated sampled-candidate protocol, datasets, and link-retention ratios; they do not establish elimination of popularity bias or complete independence from overlapping users. PACL-CDR also incurs additional model size and training time, although its measured inference time remains comparable to several representative baselines. Future work will examine full-catalog exposure, broader overlap regimes, and more efficient training for practical multi-domain deployment.

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